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No-hair theorem for minimally coupled matter with an eye for non-minimal coupling

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In this study, we examine the stability and properties of scalar fields within the exterior of Schwarzschild black holes, constrained to the context of General Relativity. This work explores the implications of the "no-hair" conjecture by analyzing the behavior of massive and massless scalar fields, along with their interactions with electromagnetic fields through minimal coupling. Utilizing a comprehensive formalism, we address the equations of motion, derive the stress-energy tensor, and discuss boundedness constraints in the near-horizon region. Through a sequence of calculations, we establish that scalar fields cannot exist in a stable form within the black hole exterior under specified assumptions, thereby reinforcing the no-hair conjecture. Our findings contribute further insight into scalar field behavior in black hole physics without extending to modified gravity frameworks.

I. INTRODUCTION

In 1915, months after Albert Einstein published his general theory of relativity, Karl Schwarzschild used his field equations to arrive at vacuum solution. This solution had peculiar properties. For instance, light wave emitted at event horizon would get infinitely redshifted, and time would slow down the closer someone got to event horizon. Very soon, it was established that stars with mass greater than certain limit would collapse under their own gravity to form these objects, which came to be known as black holes.

Black hole is a region of spacetime separated from exterior universe via one way membrane called event horizon, such that no interior particles or light rays can ever escape to the exterior. This does not mean that an external observer has no information about a black hole's interior, as any external field associated with black hole can tell what is inside. For example, an electromagnetic field in exterior of charged black hole allows an outside observer to learn about the charge it has.

It soon became a matter of interest among black hole physicists to study what other quantities of black holes can one know from its outside. Let's assume a simple system viz. stationary black holes. These are time-independent ones with no matter or radiation around them. General relativity predicts a family of black holes satisfying such exterior. These are parametrized by only mass, charge and angular momentum- a solution known as Kerr-Newman type. Israel-Carter conjecture states that all stationary, single black holes are of Kerr-Newman type: They possess no other independent property than mass, charge and angular momentum. This conjecture can be famously rephrased as "black holes have no hair."

Recent research has shown that black holes do have hair. One has to go beyond Einstein's relativity and enter

modified gravity models to study such effects.

In this paper, however, we restrict ourselves to general relativity and explore scalar fields in the exterior of simplest solutions- schwarzschild blackholes. We first make relevant assumptions, introduce the formalism [1] to work with, and then study scalar fields (massive, massless, charged with minimum coupling to electromagnetic field tensor). We have tried to show each calculation in detail.

II. ASSUMPTIONS

We attend to the situation when all matter and radiation around black hole has either been absorbed or ejected out, leaving only fields intrinsic to black hole. At this point, our black hole has settled into a stationary state. It is assumed that no naked singularities are present in the exterior to black hole, in consensus with Cosmic censorship hypothesis. Since blackholes are localised objects, exterior of BH is assumed asymptotically flat. Black hole singularity is isolated from rest of the universe via non singular, closed one way membrane, viz. horizon.

We work in units where $G = c = \hbar = 1$. The only dimension to work with is length(which is measured in units of planck length.) All fields under consideration are sourceless. No interaction is assumed between various fields, except electromagnetic coupling of charged fields with one another and with BH's exterior electromagnetic field. In passing to curved space, we use $\partial_\mu \rightarrow \nabla_\mu$ rule to obtain generally covariant form of these equations.

III. FORMALISM

We choose coordinates x^μ to cover black hole exterior, metric $g_{\mu\nu}$ which is independent of time coordinate

x^0 and asymptotically reduces to metric tensor of flat spacetime. signature of metric used is $(-+++)$. stationary horizon is described by equation $F(x^i) = 0$, where F is some function. Normal is given by $n_\mu = \partial_\mu F$. We note that normal and surface element (which is proportional to normal) have zero time components.

Let us take a set of general fields Φ_k which are associated with black hole exterior. These fields are chosen to be described by lagrangian density $\mathcal{L}$. Applying variation to action corresponding to this lagrangian, we have,

$$\begin{aligned} \delta \int \sqrt{-g} \mathcal{L} d^4x &= 0 \\ \Rightarrow \int \frac{\partial}{\partial \Phi_k} (\sqrt{-g} \mathcal{L}) \delta \Phi_k + \frac{\partial}{\partial (\partial_\mu \Phi_k)} (\sqrt{-g} \mathcal{L}) \delta (\partial_\mu \Phi_k) &= 0 \\ \Rightarrow \int \left(\left(\frac{\partial}{\partial \Phi_k} (\sqrt{-g} \mathcal{L}) \delta \Phi_k + \partial_\mu \left(\frac{\partial}{\partial (\partial_\mu \Phi_k)} (\sqrt{-g} \mathcal{L}) \delta \Phi_k \right) \right) \right. \\ \left. - \partial_\mu \left(\frac{\partial}{\partial (\partial_\mu \Phi_k)} (\sqrt{-g} \mathcal{L}) \right) \delta \Phi_k d^4x \right) &= 0 \end{aligned}$$

Second term in above expression is a total derivative and vanishes as variation of fields at endpoints is zero.

$$\begin{aligned} \Rightarrow \int \left(\left[\frac{\partial}{\partial \Phi_k} (\sqrt{-g} \mathcal{L}) - \right. \right. \\ \left. \left. \partial_\mu \left(\frac{\partial}{\partial (\partial_\mu \Phi_k)} (\sqrt{-g} \mathcal{L}) \right) \right] \delta \Phi_k d^4x \right) &= 0 \end{aligned}$$

Since the variation of fields is arbitrary, the integrand is zero for each set of fields denoted by k .

$$\Rightarrow \frac{\partial \mathcal{L}}{\partial \Phi_k} - \frac{1}{\sqrt{-g}} \partial_\mu \left(\sqrt{-g} \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi_k)} \right) = 0 \quad (1)$$

Thus we obtain the Euler-Lagrange equations. Now multiply each by $\Phi_k \sqrt{-g} d^4x$ and integrate over black hole exterior

$$\begin{aligned} \int_{\text{BH interior}} \left(\Phi_k \sqrt{-g} d^4x \left(\frac{\partial \mathcal{L}}{\partial \Phi_k} \right. \right. \\ \left. \left. - \frac{1}{\sqrt{-g}} \partial_\mu \left(\sqrt{-g} \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi_k)} \right) \right) \right) &= 0 \\ \Rightarrow \int d^4x \left(\left(\sqrt{-g} \Phi_k \frac{\partial \mathcal{L}}{\partial \Phi_k} + \partial_\mu \Phi_k \left(\sqrt{-g} \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi_k)} \right) \right) - \right. \\ \left. \int d^4x \partial_\mu \left(\Phi_k \sqrt{-g} \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi_k)} \right) \right) &= 0 \end{aligned}$$

The third term here is a 4- dimensional volume integral of total derivative. We can convert it to surface integral using gauss divergence theorem, as shown in next line:

$$\int d^4x \partial_\mu \left(\Phi_k \sqrt{-g} \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi_k)} \right) = \int \Phi_k \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi_k)} dS_\mu$$

If we sum over each field present, we get

$$\sum_k \int \sqrt{-g} d^4x \left(\Phi_k \frac{\partial \mathcal{L}}{\partial \Phi_k} + \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi_k)} \right) - \int b^\mu dS_\mu = 0 \quad (2)$$

where

$$b^\mu = \sum_k \Phi_k \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi_k)} \quad (3)$$

dS_μ is a hypersurface bounding event horizon, spatial infinity and future and past timelike infinity. Since we are focusing on static blackholes, we can put g_{0i} and g^{0i} to zero. The fact that horizon is null can then be expressed as

$$dS^\mu dS_\mu = 0 \Rightarrow g_{00} dS^0 dS^0 - 2g_{0i} dS^0 dS^i - g_{ij} dS^i dS^j = 0$$

It is known that $g_{00} = 0$ on horizon. Thus we have

$$g_{ij} dS^i dS^j = 0 \quad (4)$$

We can similarly write

$$b^\mu dS_\mu = g_{\mu\nu} b^\mu dS^\nu = \left(g_{00} b^0 dS^0 + g_{0i} b^0 dS^i + g_{i0} b^i dS^0 + g_{ij} b^i dS^j \right)$$

Using reasoning as above, we can write

$$g_{ij} b^i dS^j = 0$$

spatial metric g_{ij} is positive definite in blackhole exterior and positive semi-definite on horizon. Thus spatial distance dl between two points separated by coordinate intervals dx^i is well defined:

$$d\ell^2 = g_{ij} dx^i dx^j$$

For any arbitrary real number λ and real triplets dS^i

$$g_{ij} (dS^i + \lambda b^i) (dS^j + \lambda b^j) \geq 0$$

if we choose $\lambda = -\frac{g_{ij} dS^i b^j}{g_{ij} b^i b^j}$ (where denominator is non zero), we arrive at a Schwarz inequality as shown.

$$g_{ij} dS^i dS^j + \lambda g_{ij} dS^i b^j + \lambda g_{ij} b^i dS^j + \lambda^2 g_{ij} b^i b^j \geq 0$$

$$\begin{aligned}
&\Rightarrow g_{ij} dS^i dS^j + 2\lambda g_{ij} dS^i b^j + \lambda^2 g_{ij} b^i b^j \geq 0 \\
&\Rightarrow g_{ij} dS^i dS^j - \frac{(g_{ij} dS^i b^j)^2}{g_{ij} b^i b^j} \geq 0 \\
&\Rightarrow g_{ij} dS^i dS^j \geq \frac{(g_{ij} dS^i b^j)^2}{g_{ij} b^i b^j} \\
&\Rightarrow (g_{ij} dS^i b^j)^2 \leq (g_{ij} dS^i dS^j) (g_{lm} b^l b^m) \quad (5)
\end{aligned}$$

In all cases we will look at, it will turn out that $b^\mu b_\mu$ is bounded and $b^0 = 0$. This implies that $g_{ij} b^i b^j$ is bounded and using above equations, it is deduced that

$$dS^\mu b_\mu = 0$$

on horizon.

It will be verified later that for all physically relevant fields, b^μ vanish asymptotically. Thus no contribution is obtained from spatial infinity part to the integral. At timelike infinity, the boundary has $n_i = dS_i = 0$ so that $dS^\mu b_\mu = 0$ on it.

Combining all these results, we arrive at the conclusion, that

$$\int b^\mu dS_\mu = 0 \quad (6)$$

Thus first integral in equation (2) vanishes by itself. We will use this result to eliminate scalar fields outside a static black hole.

IV. MASSIVE SCALAR FIELDS

Now consider a real scalar field ψ with reciprocal length dimension $m > 0$. Lagrangian density used to describe it is

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \psi \partial^\mu \psi - \frac{1}{2} m^2 \psi^2 \quad (7)$$

Using Euler Lagrange equations, we find the associated equation of motion for this field.

$$\frac{\partial \mathcal{L}}{\partial \psi} = -m^2 \psi, \quad \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \psi)} \right) = \partial_\mu (-\partial^\mu \psi) = -\partial_\mu \partial^\mu \psi$$

Thus we obtain Klein-Gordon equation as shown:

$$\partial_\mu \partial^\mu \psi - m^2 \psi = 0 \quad (8)$$

We find Stress-Energy tensor using Noether's theorem. For this, we first perform spacetime translations and note how the coordinates, fields and lagrangian change.

$$x^\nu \rightarrow x^\nu + \epsilon^\nu, \quad \psi \rightarrow \psi + \epsilon^\nu \partial_\nu \psi$$

$$\delta \psi = \epsilon^\nu \partial_\nu \psi, \quad \delta \mathcal{L} = \epsilon^\nu \partial_\nu \mathcal{L}$$

We know that for considered transformation, action remains invariant. Thus, it is a symmetry of action. And Noether's theorem claims that such symmetry leads to conserved current, given by

$$J^\mu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \psi)} \delta \psi - \delta \mathcal{L}$$

Putting in relevant terms, and raising both indices, we arrive at stress energy tensor:

$$T^{\mu\nu} = \partial^\mu \psi \partial^\nu \psi - g^{\mu\nu} \mathcal{L} \quad (9)$$

Static case implies that $T^{\mu\nu}$ will also be static, so that $\partial_0 \psi = 0$. From formalism, we can write $b_\mu = -\psi \partial_\mu \psi$, so $b^0 = 0$

We want to show that $b^\mu b_\mu$ is bounded on horizon. It can be easily seen that $b^\mu b_\mu = \psi^2 \partial^\mu \psi \partial_\mu \psi$. We re-express the terms in this expression using our derived stress tensor.

Trace of $T^{\mu\nu}$ or simply, T can be written as

$$\begin{aligned}
T &= T^\mu_\mu = g_{\mu\nu} T^{\mu\nu} = \\
&g_{\mu\nu} \left(\partial^\mu \psi \partial^\nu \psi - \frac{1}{2} g^{\mu\nu} (\partial_\alpha \psi \partial^\alpha \psi + m^2 \psi^2) \right) \\
&= \partial_\mu \psi \partial^\mu \psi - \frac{1}{2} 4 (\partial_\alpha \psi \partial^\alpha \psi + m^2 \psi^2)
\end{aligned}$$

$$T = -\partial_\mu \psi \partial^\mu \psi - 2m^2 \psi^2$$

And inner product of $T^{\mu\nu}$ with itself is given as

$$\begin{aligned}
T_{\mu\nu} T^{\mu\nu} &= \left(\partial_\mu \psi \partial_\nu \psi - \frac{1}{2} g_{\mu\nu} (\partial_\alpha \psi \partial^\alpha \psi + m^2 \psi^2) \right) \\
&\quad \left(\partial^\mu \psi \partial^\nu \psi - \frac{1}{2} g^{\mu\nu} (\partial_\beta \psi \partial^\beta \psi + m^2 \psi^2) \right) \\
&= \frac{3}{4} (\partial_\mu \psi \partial^\mu \psi)^2 + \frac{1}{4} (\partial_\alpha \psi \partial^\alpha \psi + 2m^2 \psi^2)^2
\end{aligned}$$

$$T_{\mu\nu} T^{\mu\nu} = \frac{3}{4} (\partial_\mu \psi \partial^\mu \psi)^2 + \frac{1}{4} T^2$$

Thus $\partial_\mu \psi \partial^\mu \psi$ and ψ^2 can be written in terms of $T_{\mu\nu} T^{\mu\nu}$ and T , both of which are physical scalars:

$$\Rightarrow \partial_\mu \psi \partial^\mu \psi = \left(\frac{4}{3} T_{\mu\nu} T^{\mu\nu} - \frac{1}{3} T^2 \right)^{\frac{1}{2}} \quad (10)$$

$$2m^2 \psi^2 = -\partial_\mu \psi \partial^\mu \psi - T$$

$$m^2\psi^2 = -\frac{1}{2}\left(\frac{4}{3}T_{\mu\nu}T^{\mu\nu} - \frac{1}{3}T^2\right)^{\frac{1}{2}} - \frac{1}{2}T \quad (11)$$

The reason for such manipulation is simple to understand:

Suppose there is a freely falling observer. While crossing the horizon, he should find that physical quantities like j_μ , $T_{\mu\nu}$ and $F_{\mu\nu}$ remain bounded (meaning non-zero finite) unless there's something singular in the physics at horizon. However, we have excluded that possibility by considering a "non-singular" horizon, which means that local physics is nonsingular. Any scalar constructed out of these quantities will also be bounded. So this way, since $T_{\mu\nu}T^{\mu\nu}$ and T are bounded, so should be $b_\mu b^\mu$

Hence our formalism tells us that first integral in equation (1) must vanish, or

$$\int (g_{ij}\partial^i\psi\partial^j\psi + m^2\psi^2)(-g)^{1/2}d^4x = 0 \quad (12)$$

Since g_{ij} is positive definite, the only way for the integral to vanish is when ψ is identically zero throughout the black hole exterior.

V. MASSLESS SCALAR FIELD

Herein lies a subtlety. Since $m = 0$, equation (11) doesn't ensure that ψ^2 is bounded and thus we may not be able to apply the reasoning so mentioned for massive scalar field. Also, in equation (8) we note that ψ is determined only upto an additive constant. Now, we may assume that ψ vanishes asymptotically and this way ψ^2 may be interpreted as invariant probability. It then follows that since probability is a physical scalar, it is bounded on horizon. Also, the additive constant need not be zero, as it appears nowhere in physical quantity and thus is an unobservable.

VI. COMPLEX SCALAR FIELD COUPLED MINIMALLY TO ELECTROMAGNETIC FIELD

Now we take up a charged (complex) scalar field ψ which is minimally coupled to electromagnetic field by a vector potential A_μ and field tensor $F_{\mu\nu}$. Lagrangian density is given by

$$\mathcal{L} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} - (d_\alpha d^{\alpha*} + m^2\psi\psi^*) \quad (13)$$

where $d_\alpha = \partial_\alpha - ieA_\alpha\psi$. We find stress energy tensor using Noether's theorem, which tells us about the conserved current we get for symmetry in translational invariance of action:

$$J^\mu_\nu = \frac{\partial\mathcal{L}}{\partial_\mu\psi}\delta\psi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi^*)}\delta(\partial_\mu\psi^*) - \delta\mathcal{L}$$

On solving, we get required tensor

$$T^{\mu\nu} = d^{\mu*}d^\nu\psi + d^\mu d^{\nu*}\psi^* - g^{\mu\nu}(d_\alpha^*d^\alpha + m^2\psi\psi^*) \quad (14)$$

The theory is invariant under gauge transformation:

$$\psi \rightarrow \psi e^{ie\Lambda}, \quad A_\mu \rightarrow A_\mu + \partial_\mu\Lambda \quad (15)$$

which gives us another conserved current, (which has been derived using another method). From equations of motion for ψ and ψ^* , we get

$$\frac{\partial\mathcal{L}}{\partial\psi} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\right) = 0$$

$$\Rightarrow (-m^2\psi^* + ieA^\alpha\partial_\alpha\psi^* - e^2A^\alpha A_\alpha\psi^*) - \partial_\mu(-\partial_\mu\psi^* - ieA^\mu\psi^*) = 0$$

$$\Rightarrow \partial_\mu^2\psi^* - ie\partial_\mu A^\mu\psi^* + 2ieA_\mu\partial^\mu\psi^* - e^2A_\alpha A^\alpha\psi^* - m^2\psi^* = 0$$

$$\frac{\partial\mathcal{L}}{\partial\psi^*} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi^*)}\right) = 0$$

$$\Rightarrow \partial_\mu^2\psi - ie\partial_\mu A^\mu\psi - 2ie\partial^\mu\psi A_\mu - e^2A_\alpha A^\alpha\psi - m^2\psi = 0$$

Manipulating these resulting equations, we can write the final term -as derivative of a quantity -to be equal to zero.

$$\partial_\mu[\psi\partial^\mu\psi^* - \psi^*\partial^\mu\psi]ie = 0$$

The conserved electric current is thus,

$$\Rightarrow J_\mu = ie(\psi\partial_\mu\psi^* - \psi^*\partial_\mu\psi) \quad (16)$$

Since we have assumed a static situation, it is always correct to consider $J^i = 0, \partial_0 J^0 = 0$. Also we can choose $A_i = 0, \partial_0 A_0 = 0$. It thus follows that $\psi\psi^*$ is independent of coordinate x^0 , and phase of ψ must be of the form $\omega x^0 + \phi$, that is,

$$\psi = e^{i(\omega x^0 + \phi)}\psi_{static}$$

where ω and ϕ are real constants. If we choose $\Lambda = -\frac{1}{e}(\omega x^0 + \phi)$, ψ becomes real and time independent without altering the conditions on A_μ .

Now, we move on to find b_μ .

$$b_\mu = \psi\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)} + \psi^*\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi^*)}$$

$$\begin{aligned}
&= \psi (-\partial_\mu \psi^* - ieA_\mu \psi^*) + \psi^* (-\partial_\mu \psi + ieA_\mu \psi) \\
&= -\psi^* d_\mu - \psi d_\mu^*
\end{aligned}$$

It can be shown that $b_0 = 0$ and all b_i are real:

$$b_0 = \psi d_0 \psi^* + \psi^* d_0 \psi = -[\psi \partial_0 \psi^* + \psi^* \partial_0 \psi]$$

$$= -\partial_0(\psi \psi^*) = 0$$

$$b_i = -\partial_i \psi \psi^* = -\partial_i |\psi|_{\text{static}}^2$$

To show that $b_\mu b^\mu$ is bounded on horizon, we obtain following relations from stress energy tensor:

$$T = T_\mu^\mu = d^{\mu*} d_\mu + d^\mu d_\mu^* - g^{\mu\nu} (d^\alpha d_\alpha^* + m^2 \psi \psi^*)$$

$$T = -2d^\alpha d_\alpha^* - 4m^2 \psi \psi^* \quad (17)$$

$$T^{\mu\nu} T_{\mu\nu} = 2|d^\alpha d_\alpha^*|^2 + \frac{3}{4} T d^{\mu*} d_\mu + \frac{T^2}{16} + \frac{1}{2} (d^\mu d_\mu^*)^2 \quad (18)$$

As both $T^{\mu\nu}$ and T are physical scalars, they are bounded on horizon. From equation (18), terms $d^\mu d_\mu$ and $d^{\mu*} d_\mu^*$ must be bounded, otherwise anyone's unboundedness would result into $T^{\mu\nu} T_{\mu\nu}$ being unbounded. Thus in equation (17), $m^2 \psi \psi^*$ must also be bounded.

calculating $b^\mu b_\mu$:

$$b^\mu b_\mu = (\psi d^{\mu*} + \psi^* d^\mu)(\psi d_\mu^* + \psi^* d_\mu)$$

$$= \psi^2 d^{\mu*} d_\mu^* + \psi \psi^* d_\mu d^{\mu*} + \psi^* \psi d_\mu d^{\mu*} + (\psi^*)^2 d_\mu d^{\mu*}$$

We recall that for our choice of gauge, ψ is real.

$$= \psi \psi^* (d^{\mu*} d_\mu^* + d_\mu d^\mu + 2d^\mu d_\mu^*)$$

It indeed follows that $b^\mu b_\mu$ is bounded, on the horizon. We have shown that $b_0 = 0$ and $b^\mu b_\mu$ is bounded. Now we can proceed to write first integral in equation (1) which is now equal to zero. Or,

$$\int [g_{ij} \partial^i \psi \partial^j \psi + (m^2 + g_{00}(eA^0)^2) \psi^2] \sqrt{-g} d^4 x = 0 \quad (19)$$

g_{ij} is positive definite in BH exterior, but the integral isn't unless A_0 is not equal to zero.

Now we show how A_0 will be zero. From equation (16), for our gauge, we have $J^0 = -2e^2 A^0 \psi^2$. We interpret ψ^2 as invariant probability density and thus it is bounded on horizon. Since $(-J^\mu J_\mu)^{1/2}$ is charge density, we see that $\frac{(-J^\mu J_\mu)^{1/2}}{\psi^2}$ is specific charge of the field, and is a physical scalar, which must be bounded on horizon. Since $J_i = 0$, we conclude that $g_{00} A^0$ is bounded.

Let us take a look at first term in equation (13). $F^{\mu\nu} F_{\mu\nu}$ is a physical scalar. With considered gauge, $F^{\mu\nu} F_{\mu\nu} = g_{00} g_{ij} F^{0i} F^{0j}$. Calculating b_μ for electromagnetic field:

$$b^\mu = A_\nu \frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} = -A_\nu F^{\mu\nu}$$

$$b^\mu = - (A_0 F^{\mu 0} + A_i F^{\mu i}) = -A_0 F^{\mu 0}$$

Thus we find that $b^0 = 0$ and $b^\mu b_\mu = A_0 A^0 F^{\mu 0} F_{\mu 0}$ which is bounded, since we showed boundedness for A^0 while $F_{\mu\nu}$ is a physical scalar.

Formalism says that first integral of equation (2) vanishes. That is,

$$\begin{aligned}
&\int \left(A_\nu \frac{\partial \mathcal{L}}{\partial A_\nu} + (\partial_\mu A_\nu) \frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} \right) \sqrt{-g} d^4 x \\
&= \int [A_\nu (2e^2 \psi^2 A^\nu) + F_{\mu\nu} F^{\mu\nu}] \sqrt{-g} d^4 x \\
&= \int g_{00} [2e^2 \psi^2 (A^0)^2 + g_{ij} F^{0i} F^{0j}] \sqrt{-g} d^4 x
\end{aligned}$$

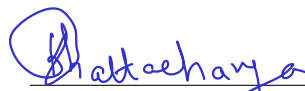
It is clear from above expression that, since g_{00} is negative in exterior region, the integral is negative definite, and it can only vanish if A^0 is identically zero everywhere outside horizon.

With this knowledge, we get back to equation (19), which has now become positive definite. Thus, the integral can only be zero if ψ vanishes throughout the exterior.

VII. CONCLUSION

This analysis reinforces the tenets of the no-hair theorem, demonstrating that scalar fields, whether massive or massless, are inherently unstable within the Schwarzschild black hole exterior under conventional assumptions of General Relativity. By rigorously deriving and applying the Euler-Lagrange equations and stress-energy tensors, we conclude that scalar fields must vanish identically to maintain stability at the horizon boundary. The results confirm that the presence of scalar hair is untenable in these conditions, further supporting the classical perspective that black holes are characterized solely by mass, charge, and angular momentum. Future work will introduce minimally coupling between with scalar fields and matter to see whether results are affected, and we will also look at non-minimal coupling affairs.

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- [1] Jacob D. Beckenstein, *Nonexistence of Baryon Number for Static Black Holes*, Phys. Rev. D 5, 1239 (15 March 1972)
- [2] C. V. Vishveshwara, *Generalization of the "Schwarzschild Surface" to Arbitrary Static and Stationary Metrics*, J. Math. Phys. 9, 1319–1322 (1968)
- [3] B. Carter, *Global Structure of the Kerr Family of Gravitational Fields*, Phys. Rev. 174, 1559 (1968)
- [4] Charles W. Misner, Kip Thorne, John Archibald Wheeler, *Gravitation*, (1973)



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